

Develop a Student Dropout Prediction Model Using Classification Techniques

II. Introduction

Student dropout prediction is an important application of machine learning in education. It predicts whether a student is likely to drop out based on attendance, marks, study hours and related factors. Institutions can use these predictions to identify at-risk students early and provide counseling or support.

III. Architecture Diagram / Flowchart

Student Dataset -> Data Preprocessing -> Feature Selection -> Train-Test Split -> Classification Model Training (Decision Tree) -> Model Evaluation -> Predict Student Status (Dropout / Not Dropout)

IV. Implementation - Tools and Technologies

Programming Language: Python
Environment: Google Colab/Jupyter
Libraries: Pandas, NumPy, Scikit-learn, Matplotlib
Algorithm: Decision Tree Classifier
OS: Windows/Linux/macOS

IV. Implementation - Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report

# Sample dataset
data={'Attendance':[90,65,80,50,95,40,70,85,55,60], 'Marks':[85,45,70,35,90,30,65,88,40,55],
      'StudyHours':[5,2,4,1,6,1,3,5,2,2], 'Dropout':[0,1,0,1,0,1,0,0,1,1]}
df=pd.DataFrame(data)
X=df[['Attendance','Marks','StudyHours']]
y=df['Dropout']
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
model=DecisionTreeClassifier(random_state=42)
model.fit(X_train,y_train)
y_pred=model.predict(X_test)
print('Accuracy:',accuracy_score(y_test,y_pred))
print(classification_report(y_test,y_pred))
new_student=[[75,68,3]]
print('Prediction:', 'Dropout' if model.predict(new_student)[0] else 'Not Dropout')
```

V. Result

Run the code in Google Colab. Example output includes model accuracy, classification report, and a prediction showing whether a new student is likely to drop out.

Explanation of Results

The Decision Tree model predicts student dropout using attendance, marks, and study hours. Lower attendance and marks generally increase dropout risk.

Runnable Project Link

After running in Google Colab, click Share > Anyone with the link > Copy Link.

Output Screenshot

.. Accuracy: 1.0

```
Classification Report
              precision    recall  f1-score   support

         1         1.00      1.00      1.00         2

 accuracy          1.00
macro avg          1.00      1.00      1.00         2
weighted avg          1.00      1.00      1.00         2
```

Prediction: Student is Not likely to Dropout

Updated Runnable Project Link

<https://colab.research.google.com/drive/1cO13C3UWMgm3HJgeFOBxv8tsC7NFRkLY?usp=sharing>

Github Link: <https://github.com/ommalewar8154-wq/A-Student-Dropout-Prediction-Model>

VI. Viva Questions

1. What is student dropout prediction? - Predicting whether a student will leave before completion.
2. Which ML technique is used? - Classification.
3. Which algorithm is used? - Decision Tree.
4. Why use train-test split? - To evaluate model performance.
5. What is accuracy? - Percentage of correct predictions.